Engine fault detection from sound

Can a short recording help a fleet prioritize inspections?

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Engine inspection triage

A service writer records up to 10 seconds of engine sound.
A fault flag helps a technician decide what to inspect first.

≥85%

Mean recall across
knock and misfire

≤20%

Healthy clips
flagged for inspection

<1 s

Target per clip
on a laptop CPU

Missed faults can delay repairs. False alarms use technician time.

The KweriMoto recordings

1,067 clips, three labels, up to 10 seconds each

Recording source	Healthy	Knock	Misfire
AudioSet	0	322 real	0
Revix	271	286 synthetic	0
Hugging Face	80	0	108

721 train, 154 validation, 154 mixed test, 38 real-knock test

We kept related recordings and overlay sounds in the same split.

A pre-trained model with a small classifier

Audio Spectrogram Transformer (AST)

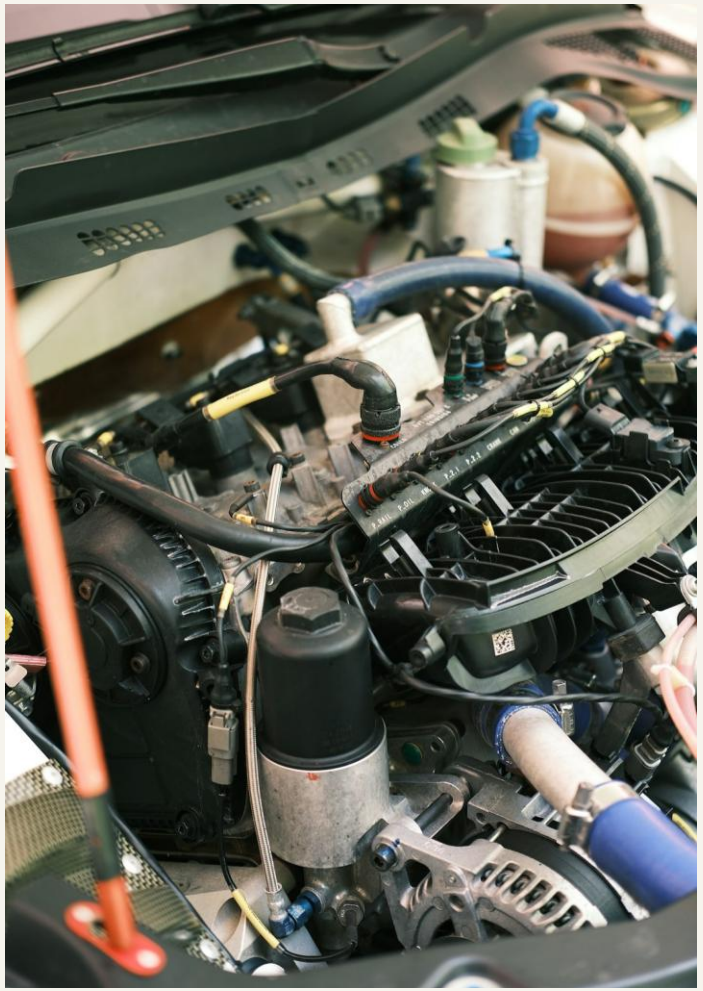
AudioSet training gives AST a starting point for recognizing patterns in engine sounds.

Frozen AST baseline

AST summarizes each recording. We train a 2,307-parameter classifier on those summaries.

Alternative model

CLAP matches audio to text prompts. Its zero-shot fault recall was 22.2% on validation.



Validation results across the five runs

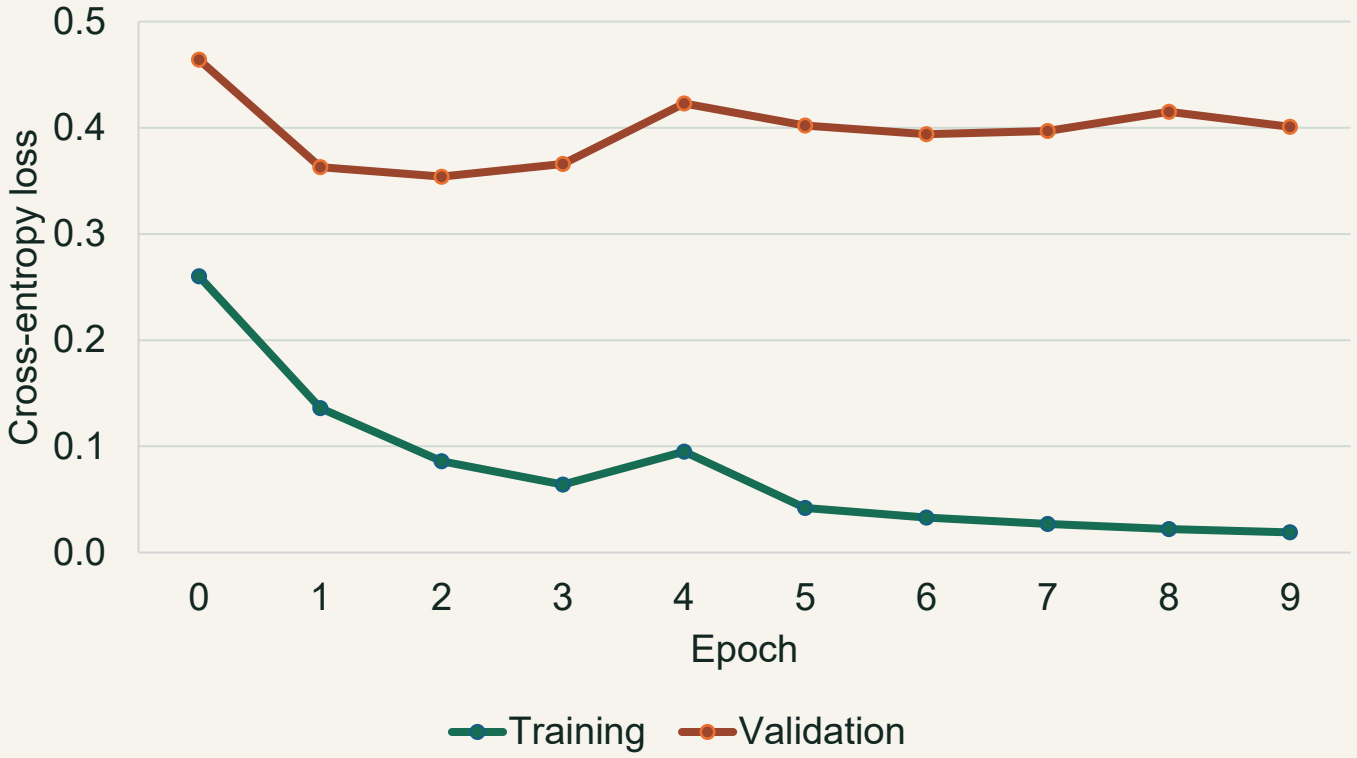
Same grouped validation set for every run. Targets were ≥85% recall and ≤20% false alarms.

Run	Change	Fault recall	False alarms
Benchmark	Always predict knock	50.0%	100.0%
Baseline	Frozen AST + linear classifier	81.5%	19.2%
1. Fine-tune	Train the top two AST blocks	80.5%	21.2%
2. Class weights	**Increase weight for rare faults**	**86.0%**	**19.2%**
3. Real knocks only	Remove synthetic training knocks	77.5%	15.4%

Class weighting recovered two misfires and missed three additional knocks.

Fine-tuning missed more misfires

Validation loss reached its minimum at epoch 2.



Misfires caught

14 of 16 with the baseline
12 of 16 after fine-tuning

Training cost

3 minutes to train
Baseline took about 2 seconds

Three confident validation errors

Baseline predictions

Actual recording	Predicted label	Confidence
Revix synthetic knock, 4.1 s	Healthy	96.8%
AudioSet real knock, 10 s	Healthy	95.3%
Hugging Face healthy engine	Misfire	92.2%

Engine noise could mask an overlay. Even a full-length real knock was missed.

A high model score can still delay an inspection or send a healthy vehicle to the shop.

AST features reveal recording source

98.1%

Source-classifier accuracy
on healthy validation clips

76.9%

Accuracy from guessing
the most common source

Fault signals still appeared within a source.

Baseline AUC was 0.865 for Revix knock overlays and 0.859 for Hugging Face misfires.

Class weights helped across seeds

Ten paired training seeds on the same validation clips

Configuration	Mean fault recall	Mean false alarms	Seeds meeting both
Frozen baseline	76.5%	20.0%	0 of 10
Class-weighted head	**84.7%**	**19.6%**	**2 of 10**

+8.2 points

Average gain in fault recall. The weighted head improved recall in all ten paired runs.

Held-out performance of the selected model

Class-weighted frozen AST, chosen on validation before test evaluation

90.6%

Mean fault recall
154 mixed test clips

11.3%

Healthy false alarms
6 of 53 healthy clips

33 of 38

Real knocks caught
separate knock-only split

247 ms per clip on the test machine’s CPU for AST feature extraction.

Recommendation for the fleet

Continue development before a pilot

Collect recordings under the same conditions

Record healthy engines and confirmed faults across vehicles with the same phone and procedure.

Validate on different fleet vehicles

Check both decision targets and measure latency on the shop’s hardware.

How much recall would carry over to a different fleet?